

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
reasoning process here
de Broglie wavelength is given by $\lambda = h/p$, where h is plank's constant and p is momentum.
momentum, p , can be found from kinetic energy, K , using $p = \sqrt{2 \cdot m \cdot K}$, where m is the mass of the particle.
planks constant, h , is approximately 6.625×10^{-27} erg sec.
kinetic energy, K , is 1 eV, which is approximately 1.602×10^{-19} J.
mass of an electron, m , is approximately 9.109×10^{-28} g.
first, calculate the momentum, p .
 $p = \sqrt{2 \cdot m \cdot K} = \sqrt{2 \cdot (9.109 \times 10^{-28} \text{ g}) \cdot (1.602 \times 10^{-19} \text{ J})} = 3.008 \times 10^{-26} \text{ g m/s}$.
now, calculate the wavelength, λ .
 $\lambda = h/p = (6.625 \times 10^{-27} \text{ erg sec}) / (3.008 \times 10^{-26} \text{ g m/s}) = 2.20 \times 10^{-1} \text{ cm}$.
converting to the requested units, $\lambda = 2.20 \times 10^{-1} \text{ cm} \cdot 10^7 \text{ cm/m} = 2.20 \times 10^6 \text{ cm} = 2.20 \times 10^{-1} \text{ m}$.
now, convert to the requested units, $\lambda = 2.20 \times 10^{-1} \text{ m} \cdot 10^4 \text{ cm/m} = 220 \text{ cm}$.
now, select the closest option.
</think>
<answer>
F
</answer>